

Koszul Duality

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We give a brief outline of Koszul rings. For more details, see [1, Section 2].

1 Characterizations and First Implications

We start with some motivation. One useful feature of the symmetric algebra of a vector space over a field k (or more generally a free module over a semisimple ring k) is the Koszul complex resolving k . From this resolution, we get for example the Hilbert syzygy theorem, Mumford's regularity theorem, etc. It also hints at a relationship between the symmetric algebra and the exterior algebra (will see more on this later).

To define a Koszul ring in general, we essentially take the Koszul complex resolving k as an axiom.

Definition 1.1. A **Koszul ring** is a $\mathbb{Z}_{\geq 0}$ -graded ring A with $k = A_0$ semisimple such that there exists a *linear projective resolution* of k as a graded left A -module, i.e., a long exact sequence

$$\cdots \rightarrow P^2 \rightarrow P^1 \rightarrow P^0 \rightarrow k \rightarrow 0$$

such that $P^i = AP_i^i$ (i.e., generated in degree i).

Remark 1.2. We make a few comments.

- We apologize for breaking the (co)homology indexing convention by having descending superscripts—we do this so we can easily distinguish between homological degree (superscript) and grading degree (subscript).
- The linear projective resolution of k is unique up to unique isomorphism, since being linear forces it to be a minimal resolution.

We now give some implications/characterizations of Koszul rings. From now on, A will always indicate a $\mathbb{Z}_{\geq 0}$ -ring with $k = A_0$ semisimple. Denote hom graded Hom , and ext the derived functor of hom .

Proposition 1.3. A is Koszul iff $n \neq i$ implies $\text{ext}_A^i(k, k(n)) = 0$.

In other words, Koszul rings are “as close to semisimple as possible” in that there are very few extensions. Next, we will see that Koszul rings are a special type of quadratic algebra.

Definition 1.4. A **quadratic algebra** is a graded ring over k generated in degree 1 with relations generated in degree 2, i.e., isomorphic to

$$TM/(R)$$

where $M = A_1$ is a k -bimodule and $R \subseteq M \otimes_k M$.

Proposition 1.5. The statement $n \neq i$ implies $\text{ext}_A^i(k, k(n))$ for

- $i \leq 1$ implies A is generated in degree 1.
- $i \leq 2$ implies A is a quadratic algebra.

In particular, Koszul algebras are quadratic algebras.

Next, we can ask which quadratic algebras are Koszul. It turns out we can always define a Koszul complex, but it need not be exact.

Construction 1.6. Let $A = TV/(R)$ be a quadratic algebra. First, we define what will be the terms in a complex. Define k -bimodules K_i^i by

$$K_i^i = \bigcap_{\nu=0}^{i-2} V^{\otimes \nu} \otimes R \otimes V^{\otimes i-\nu-2} \leq V^{\otimes i}$$

and put $K^i = A \otimes_k K_i^i$. Notice for $i = 0, i = 1$, the intersection inside $V^{\otimes i}$ is indexed by an empty set, so we mean that

$$K^0 = A \otimes_k V^{\otimes 0} = A, \quad K^1 = A \otimes_k V.$$

For $i = 2$, there's only one term in the intersection, and we get $K^2 = A \otimes_k R$.

Now, we define a differential $d : K^i \rightarrow K^{i-1}$ by restricting the multiplication/contraction map

$$d : A \otimes V^{\otimes i} \rightarrow A \otimes V^{\otimes i-1}$$

to $K^i \rightarrow K^{i-1}$ (checking that $d(K^i) \subseteq K^{i-1}$). Then, $d^2 = 0$ because, roughly speaking, we baked the degree 2 R into K^i , so after applying d twice we get a full copy of R inside the tensor component of A , where R is 0. **justify this better.**

Now, the complex

$$\cdots \rightarrow K^2 \rightarrow K^1 \rightarrow K^0 \rightarrow 0$$

is called the **Koszul complex**. It need not in general be a resolution of k (i.e., an exact complex everywhere except in homological degree 0, where $\text{coker}(K^1 \rightarrow K^0) \cong k$).

Proposition 1.7. $A = TV/(R)$ is Koszul iff the Koszul complex is a resolution of k .

Proof. If the Koszul complex is a resolution of k , then we have our desired linear projective resolution for our original definition of Koszul rings.

Conversely, if A is Koszul, we can show vanishings of the homologies of the Koszul complex inductively, using ext vanishings to see that if homology at homological degree p exists, it is generated in grading degree $p + 1$, but then one can now check this by hand. $\square$

2 Duality

There's a somewhat better way of describing the Koszul complex, using the notion of a quadratic dual.

Definition 2.1. Let V be a k -module, $R \subseteq V \otimes V$. The **quadratic dual** $A^!$ to $A = TV/(R)$ is the algebra

$$TV^*/(R^\perp)$$

where $R^\perp \subseteq (V \otimes V)^*$ is defined as

$$R^\perp = \{f \in (V \otimes V)^* \mid f(R) = 0\}.$$

Example 2.2. Let V be a finite dimensional k -vector space, then

$$\mathrm{Sym}^\bullet V = TV/(v_i v_j - v_j v_i)$$

and

$$\wedge^\bullet V^* = TV^*/(f_i f_j + f_j f_i)$$

are quadratic duals (check!).

Then, notice that

$$A_i^! = (V^*)^{\otimes i} / \sum_{\nu=0}^i (V^*)^{\otimes i-\nu-2} \otimes R^\perp \otimes (V^*)^{\otimes \nu},$$

so that (dualing at a right k -module since we had originally dualized V as a left k -module)

$$*(A_i^!) = \bigcap_{\nu=0}^{i-2} V^{\otimes \nu} \otimes R \otimes V^{\otimes i-\nu-2} = K_i^i,$$

so we can consider the Koszul complex as having terms

$$\cdots \rightarrow A \otimes *(A_2^!) \rightarrow A \otimes *(A_1^!) \rightarrow A \otimes *(A_0^!) \rightarrow 0$$

or maybe even better consider the terms as $\mathrm{Hom}(A_i^!, A)$ in the category of right k -modules. We can then describe the differential as the composition

$$\mathrm{Hom}(A_i^!, A) \rightarrow \mathrm{Hom}(A_i^! \otimes V, A \otimes V) \rightarrow \mathrm{Hom}(A_{i-1}^!, A)$$

where the first map is tensoring by id_V , and the second is defined by precomposing by

$$A_i^! = A_i^! \otimes k \xrightarrow{(-) \otimes \mathrm{eval}} A_i^! \otimes V^* \otimes V \xrightarrow{\text{multiplication}} A_{i+1}^! \otimes V$$

and postcomposing by multiplication $A \otimes V \rightarrow A$.

3 Linear Projective Complexes

See [2].

Definition 3.1. Let A be a $\mathbb{Z}_{\geq 0}$ -graded algebra. Denote $\mathcal{LC}_A$ the category of **linear complexes of projectives**, which has

- objects cochain complexes $(P^\bullet, d)$ with each P^k a projective graded left A -module such that P^k is generated in degree k ,
- morphisms usual chain maps.

We will assume $k = A_0$ is a field, and this will also make projective graded modules over A the same as tensor products of a graded k -vector spaces with A .

Proposition 3.2. Let A, B be quadratic dual algebras. Then,

$$\text{Gr}A\text{-Mod} \simeq \mathcal{LC}_B.$$

Proof. We define the functors forming the inverse equivalence. Let M be a graded B -module. We define a cochain complex

$$\cdots \rightarrow M_k \otimes A \rightarrow M_{k+1} \otimes A \rightarrow \cdots,$$

where the differential is the data of a map

$$M_k \rightarrow M_{k+1} \otimes A_1.$$

We get this differential by considering the multiplication map

$$M_k \otimes B_1 \rightarrow M_{k+1},$$

which we tensor with $A_1 = B_1^*$ and precompose with the map

$$M_k \otimes k \rightarrow M_k \otimes A_1 \otimes B_1$$

sending $1 \in k$ to $\text{eval} \in A_1 \otimes B_1$.

Going backwards, we take a linear projective complex (M^k, d) and define a B -module M by first defining the k -vector space

$$M = \bigoplus M^k,$$

and to define the B -module structure, we just need to give B_1 multiplication (satisfying the relations). For $b \in B_1$ and $m \in M_k$, we define

$$b \cdot m := \text{eval}_b(d(1 \otimes m))$$

since $d(1 \otimes m) \in A_1 \otimes M_{k+1}$. □

Remark 3.3. The result can be stated in a stronger way—for any graded algebra, we can associate a quadratic algebra to it, and then linear projective complexes over the original are equivalent to modules over the quadratic dual of the associated algebra.

References

- [1] Alexander Beilinson, Victor Ginzburg, and Wolfgang Soergel. Koszul duality patterns in representation theory. *Journal of the American Mathematical Society*, 9(2):473–527, 1996.
- [2] Roberto Martínez Villa and Manuel Saorín. Koszul equivalences and dualities.